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TITLE: ANALYSIS AND DESIGN OF A BUS RESERVATION SYSTEM

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I. INTRODUCTION

Transportation plays a vital role in the daily activities of individuals and organizations. However, many bus transport companies still rely on manual or semi-digital methods for ticket booking and seat allocation. These traditional systems are often inefficient, time-consuming, and prone to errors such as double booking, incorrect seat allocation, and loss of customer records.

With the advancement of technology, there is a need for a reliable and automated system that allows passengers to conveniently search for buses, select destinations, book seats, and make payments digitally. A bus reservation system helps to improve efficiency, accuracy, and customer satisfaction by simplifying the entire booking process.

II. AIM

The aim of this project is to design a **Bus Reservation System** that enables passengers to easily book bus tickets, select destinations and seats, make secure payments, and receive tickets electronically, while allowing administrators to manage buses, routes, and schedules efficiently.

III. OVERVIEW OF THE SYSTEM

The Bus Reservation System is a software-based application that allows passengers to:

- Search for available buses
- Choose departure location and destination
- Select seats
- Book tickets
- Make payments using a payment service (e.g. MTN Mobile Money)
- Generate and download tickets

Administrators can manage buses, assign drivers, and oversee the reservation process.

This system reduces human errors, minimizes delays, and provides a smooth booking experience for both passengers and transport companies.

IV. PROPOSED SDLC MODEL: THE AGILE METHODOLOGY

The Agile methodology was selected for the development of the Bus Reservation System because it supports flexibility and continuous improvement. The system is developed in short iterations (sprints), with each sprint delivering a functional part of the application such as bus search, booking, or payment.

Advantages of the Agile Methodology

- Allows easy modification of requirements
- Allows for regular testing and validation with stakeholders (passengers, bus operators).
- Encourages continuous user feedback
- Enables early delivery of working features
- Improves system quality through frequent testing

V. ANALYSIS

During the analysis phase, the functional and non-functional requirements of the system were identified.

Functional Requirements:

- User registration and login
- Search for available buses

- Select destination and seat
- Book ticket
- Make payment
- Generate and download ticket
- Cancel booking

Non-Functional Requirements:

- System should be easy to use
- Secure payment processing
- Fast response time
- Data consistency and reliability

VI. DESIGN

To design the Bus Reservation System, the **Unified Modeling Language (UML)** was used. UML diagrams help visualize the structure and behavior of the system, making it easier for stakeholders to understand how the system works.

The following diagrams were used:

- Use Case Diagram
- Class Diagram
- Sequence Diagrams
- Activity Diagrams

USE CASE DIAGRAM

The use case diagram shows the interactions between users and the system.

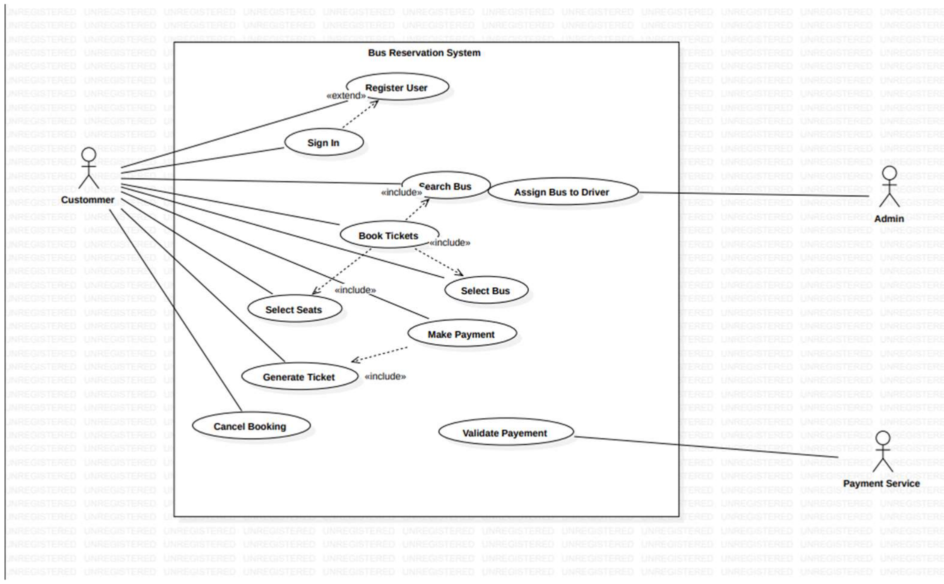
Actors:

- **Passenger** (Primary Actor)
- **Admin** (Secondary Actor)
- **Payment Service** (External System Actor)

Main Use Cases:

- Register / Sign In
- Search Bus
- Select Destination

- Select Seat
- Book Ticket
- Make Payment
- Generate Ticket
- Cancel Booking
- Assign Bus to Driver



TEXTUAL DESCRIPTION

A software system designed to allow passengers to conveniently book bus tickets by selecting destinations, seats, and making payments electronically.

1. **REGISTEER USER**

- Type: base use case
- Description: This allows a new user to register on the system. The user provides personal details and upon successful registration, the user is able to log in and access all features (e.g.
  booking tickets, viewing schedules and managing reservations)
- Business goals and benefits: Registration is essential to maintain user records, enabling personalized services, booking tracking, and secure login access. It helps the system identify and authenticate users, improving data integrity and user experience
- Actors:
  Primary actor: new user
  Secondary actor: system

- Rules of Precedence
    - Triggers: The user clicks the **Register** button on the login page of the system.
    - Pre-conditions: User does not currently have access to the system.

  User does not already have an account with the same email or phone number.

- Post-conditions

  Post-conditions on Success:
    - A new user account is created and stored in the system database.
    - User is redirected to the login page or automatically logged in.
  - Post-conditions on Failure:
    - Error message is displayed (e.g., “Email already in use,” or “Required fields missing”).

Registration form remains available for correction.

- Extension Points:
- Email verification (optional extension for verifying the user’s email after registration)
- Admin approval (if registration needs backend confirmation)

2. **SIGN IN USER**

  - Type: Base use case
  - Description: Allows already registered users to sign into their accounts and make use of the various features of the platform. These users will have already existing information of previous travels and reservations instead of creating new ones
  - Business goals and benefits: Signing is essential for accessing features, allows remote access of user information and secures user data.
  - Actors o Primary actors: Passengers - Rules of precedence:
    - Triggers: The user clicks the **Sign in** button on the login page of the system and enters his credentials.
    - Pre-conditions: User is not currently logged in.
  - Post conditions:

  Post-conditions on Success:

  - The user account is opened and his access to the system retained.
  - User is automatically logged in. o Post-conditions on Failure:
  - Error message is displayed (e.g., “Account doesn’t exist” or “Required fields missing”).
  - Extension point: None

3. **Use Case: Search Bus**
   - Type: Base use case
   - Brief Description: This use case allows the Passenger to search for available buses by entering travel criteria such as source, destination, and travel date.
   - Business Goals and Benefits: Increases user engagement and streamlines the booking process by providing quick access to relevant travel options.
   - Actors
     - Primary Actors: Passenger
     - Secondary Actors: None
     - Off-Stage Stakeholders: Bus Operator
   - Rules of Precedence
     - Triggers: Passenger accesses the search interface.
     - Pre-conditions: Passenger is logged in or accessing as a guest.
   - Post-conditions
     - Post-conditions on Success: System displays a list of available buses matching the search criteria.
     - Post-conditions on Failure: System displays an error message (e.g., "No buses found").
   - Extension Points: Filter results by price, time, or bus type.

4. Use Case: Select Bus
   - Type: Base use case
   - Brief Description: This use case allows the Passenger to choose a specific bus from the list of available options after searching.
   - Business Goals and Benefits: Enables the user to proceed with booking by selecting a preferred bus.
   - Actors
     - Primary Actors: Passenger
     - Secondary Actors: None
     - Off-Stage Stakeholders: Bus Operator
   - Rules of Precedence
     - Triggers: Passenger clicks on a bus from the search results.
     - Pre-conditions: At least one bus is available for the selected route and date.
   - Post-conditions
     - Post-conditions on Success: System displays the bus details and available seats.
     - Post-conditions on Failure: System displays a message that the bus is no longer available.
   - Extension Points: View bus amenities, see reviews.

5. Use Case: Select Seats

- Type: Base use case
- Brief Description: This use case allows the Passenger to choose one or more seats from the seating layout of the selected bus.
- Business Goals and Benefits: Gives the Passenger control over seating preference, improving customer satisfaction.
- Actors
  - Primary Actors: Passenger
  - Secondary Actors: None
  - Off-Stage Stakeholders: Bus Operator
- Rules of Precedence
  - Triggers: Passenger proceeds after selecting a bus.
  - Pre-conditions: Bus is selected and seats are available.
- Post-conditions
- Post-conditions on Success: Selected seats are temporarily reserved.
- Post-conditions on Failure: Seats are released back to available pool.
- Extension Points: View seat type (e.g., window, aisle), premium seat selection.

6. Use Case: Book Tickets

- Type: Base use case
- Brief Description: This use case allows the Passenger to confirm the booking by entering passenger details and agreeing to terms.
- Business Goals and Benefits: Converts seat selection into a confirmed booking, generating revenue.
- Actors
  - Primary Actors: Passenger
  - Secondary Actors: None
  - Off-Stage Stakeholders: Bus Operator, Admin
- Rules of Precedence
  - Triggers: Passenger clicks "Book Now" after selecting seats.
  - Pre-conditions: Seats are selected and not yet booked by another user.
- Post-conditions
  - Post-conditions on Success: A pending booking is created, and the system proceeds to payment.
  - Post-conditions on Failure: Seats are released, and booking is not created.

- Extension Points: Apply promo codes, select insurance options.

7. Use Case: Make Payment

- Type: Base use case
- Brief Description: This use case allows the Passenger to pay for the booked tickets using a preferred payment method.
- Business Goals and Benefits: Completes the transaction and finalizes the booking.
- Actors
    - Primary Actors: Passenger
    - Secondary Actors: Payment Gateway
    - Off-Stage Stakeholders: Admin, Finance System
- Rules of Precedence
    - Triggers: Passenger proceeds to payment after booking.
    - Pre-conditions: Booking is in pending status.
- Post-conditions
    - Post-conditions on Success: Payment is confirmed, booking is confirmed, and ticket is generated.
    - Post-conditions on Failure: Payment fails, booking remains pending or is cancelled.
- Extension Points: Multiple payment methods (card, wallet, UPI), installment payment.

8. Use Case: Validate Payment

- Type: Included use case (included in Make Payment)
- Brief Description: This use case validates the payment details and transaction authenticity before confirming the booking.
- Business Goals and Benefits: Ensures secure and valid transactions, reduces fraud.
- Actors
    - Primary Actors: System
    - Secondary Actors: Payment Gateway
    - Off-Stage Stakeholders: Admin
- Rules of Precedence
    - Triggers: Payment request is submitted.
    - Pre-conditions: Payment details are entered.
- Post-conditions
    - Post-conditions on Success: Payment is approved.
    - Post-conditions on Failure: Payment is declined.

- Extension Points: OTP validation, 3D Secure authentication.

9. Use Case: Generate Ticket
   - Type: Base use case
   - Brief Description: This use case generates a digital ticket (PDF/QR code) after successful payment.
   - Business Goals and Benefits: Provides proof of booking and travel authorization to the Passenger.
   - Actors
     - Primary Actors: System
     - Secondary Actors: Passenger
     - Off-Stage Stakeholders: Bus Operator
   - Rules of Precedence
     - Triggers: Payment is confirmed.
     - Pre-conditions: Booking is confirmed and paid.
   - Post-conditions
     - Post-conditions on Success: Ticket is generated and sent to Passenger.
     - Post-conditions on Failure: Notification sent to admin for manual intervention.
   - Extension Points: Send ticket via email/SMS, add to mobile wallet.

10. Use Case: Cancel Booking

   - Type: Base use case
   - Brief Description: This use case allows the Passenger to cancel a confirmed booking, subject to cancellation policy.
   - Business Goals and Benefits: Manages customer cancellations and frees up seats for rebooking.
   - Actors
     - Primary Actors: Passenger
     - Secondary Actors: None
     - Off-Stage Stakeholders: Admin, Finance System
   - Rules of Precedence
     - Triggers: Passenger selects "Cancel Booking" from booking history.
     - Pre-conditions: Booking exists and is not yet expired or departed.
   - Post-conditions
     - Post-conditions on Success: Booking is cancelled, seats are released, refund is initiated if applicable.
     - Post-conditions on Failure: Cancellation is not allowed (e.g., past cancellation window).

- Extension Points: Partial cancellation, refund processing, cancellation fee calculation.

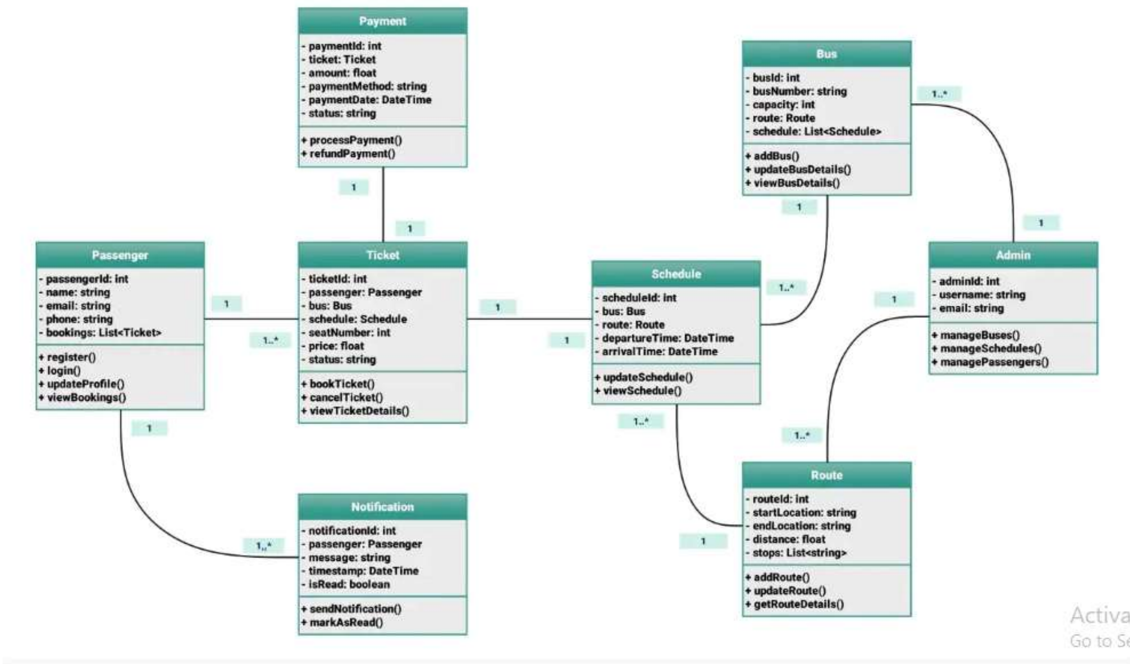
11. Use Case: Assign Bus to Driver
    - Type: Base use case
    - Brief Description: This use case allows the Admin to assign a driver to a specific bus schedule.
    - Business Goals and Benefits: Ensures proper staffing and operation of bus services.
    - Actors
        - Primary Actors: Admin
        - Secondary Actors: Driver
        - Off-Stage Stakeholders: Bus Operator
    - Rules of Precedence
        - Triggers: Admin accesses driver assignment interface.
        - Pre-conditions: Bus schedule exists, driver is available.
    - Post-conditions
        - Post-conditions on Success: Driver is assigned to the bus schedule.
        - Post-conditions on Failure: Assignment fails (e.g., driver already assigned).
    - Extension Points: View driver roster, assign alternate driver.

THE CLASS DIAGRAM

The class diagram describes the structure of the system by showing classes, their attributes, methods, and relationships.

Main Classes:

- Passenger
- Payment
- Ticket
- Schedule
- Bus
- Admin
- Notification
- Route

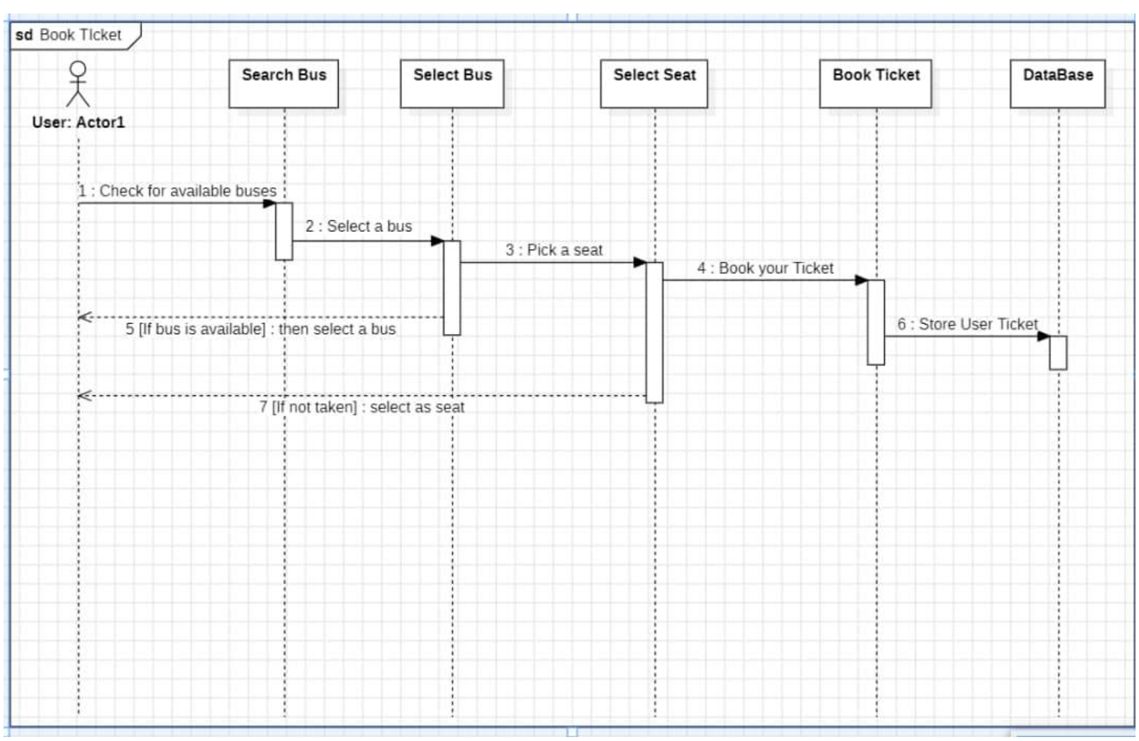


Class Diagram.

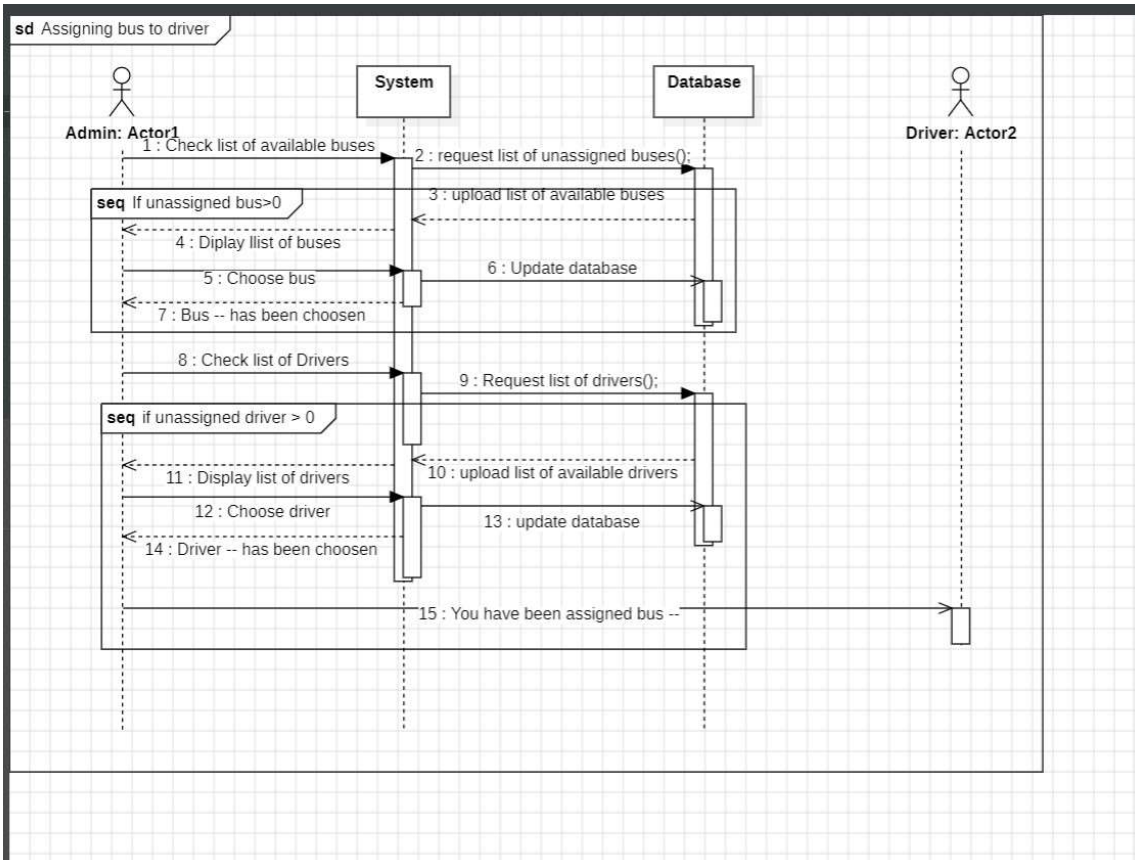
SEQUENCE DIAGRAMS

Sequence diagrams show how objects interact over time.

Sequence Diagram:



- *Book Ticket*



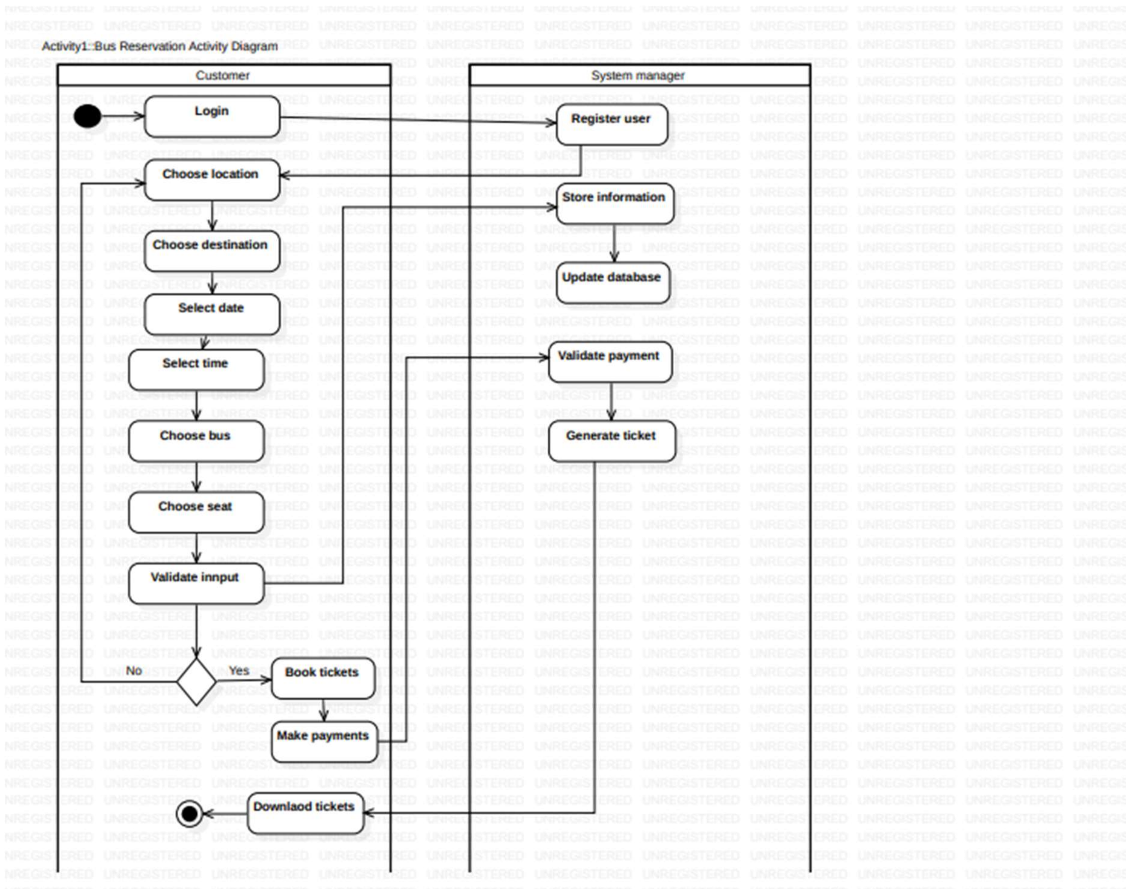
- *Assign Bus to Driver*

- *Register User*

Sequence Diagram Images Here

ACTIVITY DIAGRAM

The activity diagram illustrates the workflow of the system from ticket search to ticket generation.



Activity Diagram

VII. ADVANTAGES OF THE SYSTEM

- Reduces manual work
- Improves booking accuracy
- Saves time for passengers
- Enhances customer satisfaction
- Secure payment processing

VIII. SYSTEM CONSTRAINTS

- Requires internet connection
- Payment service availability
- Does not support offline booking

IX. FUTURE WORKS

- Seat recommendation system
- Real-time bus tracking
- Integration with multiple payment services
- SMS ticket confirmation

X. CONCLUSION

In conclusion, the Bus Reservation System provides an efficient and reliable solution for managing bus ticket bookings. By applying UML and object-oriented modeling techniques, the system design clearly represents both structural and behavioral aspects. This project has enhanced our understanding of system analysis, design, and modeling using UML.